

Technical specification

Valve-regulated lead-acid battery with Absorbent Glass Mat (AGM) separator. Sealed construction for standby power applications.

12 V

6 cells

90 Ah

10 h to 10.8 V at 25°C

400 W/cell

15 min to 1.67 V/cell at 25°C

Electrical performance

10-hour capacity	90 Ah at 9.00 A to 10.8 V, 25°C
20-hour capacity	94 Ah at 4.70 A to 10.8 V, 25°C
Internal resistance	≤ 4.4 mOhms at 25°C
Maximum discharge	810A for 5 seconds at 25°C
Self-discharge	3% average per month at 20°C
Design life / standby	12 years

Charging and operating conditions

Charge method	Constant voltage
Float / cycle voltage	13.38-13.80 V / 14.40-14.70 V at 25°C
Maximum charge current	28.2 A
Temperature compensation	Float: -20 mV/°C per battery; cycle: -30 mV/°C per battery
Operating temperatures	Discharge: -20 to +60°C; charge: -10 to +60°C
Storage temperature	-20 to +60°C

Mechanical specification

Dimensions (L × W × H)	306 × 168 × 210 mm; overall height 215 mm
Approximate weight	29.2 kg $\pm 5\%$
Terminal / torque	See terminal drawing; Not specified
Container	ABS, UL94-HB

Construction & applications

High-output AGM construction for UPS standby power. Computer-designed grids, optimized paste and lead-calcium-tin plates support high energy density. Design life: 12+ years at 20°C.

Battery construction

Positive plate	Lead dioxide
Negative plate	Lead
Container / lid	ABS / ABS
Safety valve	Rubber
Terminals	Copper
Separator	Glass fibre (AGM)
Electrolyte	Sulfuric acid

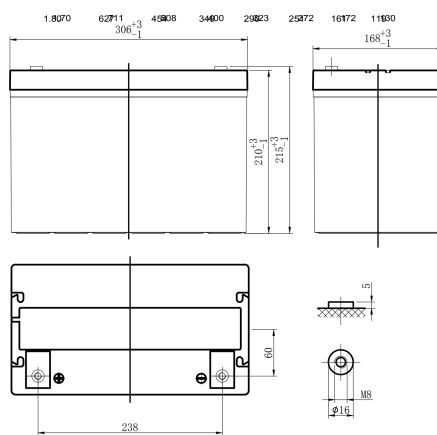
Features

- Insulated intercell connections.
- Gas recombination efficiency up to 99%.
- High-rate discharge performance.
- Special welding process.
- Self-regulating valves; no water topping-up.
- Shelf life without recharge: 6 months at 20°C.
- Low internal pressure; special AGM separators.
- Optimized for 5–15 min; loads up to 1 hour.

Applications

- Uninterruptible power supplies (UPS).
- Telecommunications backup power.
- Power and energy infrastructure.
- Security and fire alarm systems.
- Renewable-energy systems.
- Medical equipment.
- Emergency lighting.
- Data centres.

Dimensions and terminal details / mm



Dimensions and tolerances govern; do not scale the drawing.

Discharge performance

Fully charged battery / 25°C

Constant-current discharge / A

V/cell	3m	5m	10m	15m	20m	25m	30m	45m	1h
1.60	446.00	376.00	286.00	215.00	182.00	161.00	145.00	100.00	71.20
1.65	441.00	366.00	270.00	211.00	180.00	158.00	139.00	99.50	68.70
1.67	439.00	362.00	264.00	210.00	179.00	156.00	137.00	99.30	67.70
1.70	425.00	352.00	251.00	200.00	171.00	148.00	129.00	86.10	65.10
1.75	408.00	339.00	245.00	194.00	167.00	144.00	125.00	84.00	63.20
1.80	397.00	318.00	236.00	188.00	156.00	137.00	120.00	82.60	60.50

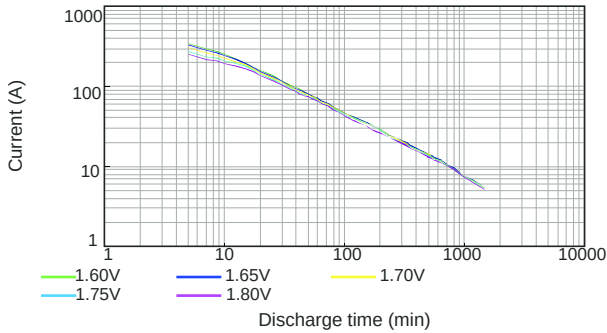
Constant-power discharge / W per cell

V/cell	3m	5m	10m	15m	20m	25m	30m	45m	1h
1.60	874.00	719.00	525.00	412.00	349.00	300.00	259.00	187.00	141.00
1.65	860.00	686.00	505.00	403.00	336.00	289.00	249.00	182.00	138.00
1.67	854.00	672.00	497.00	400.00	331.00	284.00	245.00	181.00	137.00
1.70	821.00	642.00	487.00	390.00	321.00	274.00	235.00	170.00	132.00
1.75	764.00	607.00	457.00	358.00	311.00	264.00	226.00	164.00	125.00
1.80	708.00	567.00	427.00	344.00	296.00	258.00	221.00	159.00	121.00

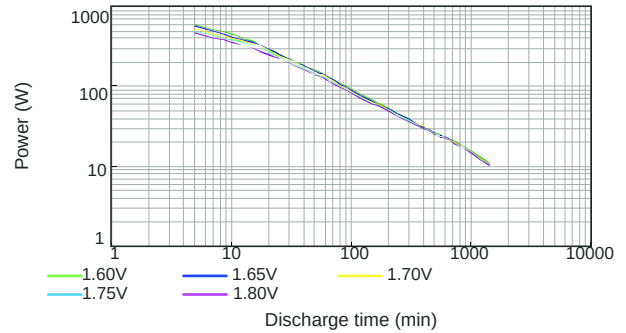
End voltage is per cell. Multiply W/cell by 6 for power per 12 V battery. Current is not divided by 6. Values are averages obtained within three charge/discharge cycles, not guaranteed minimums. Fully charge before testing. Product information may change without notice.

Characteristic curves

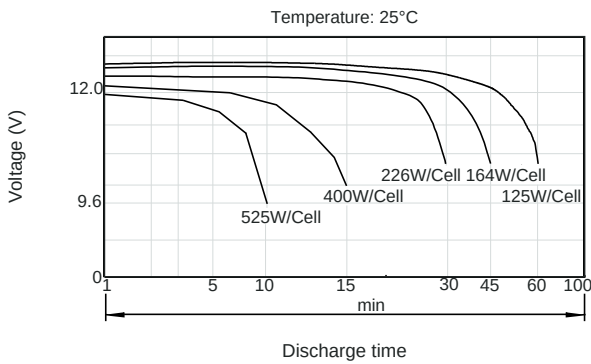
Constant-current discharge



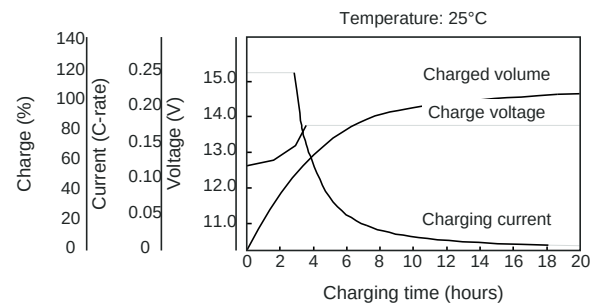
Constant-power discharge



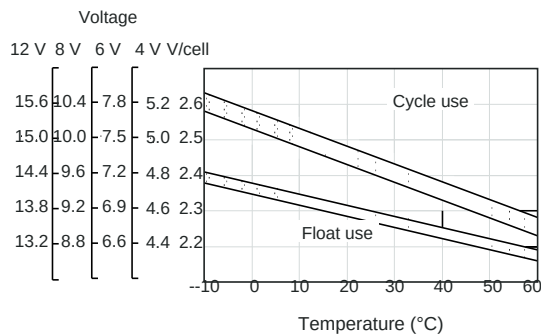
Discharge characteristics



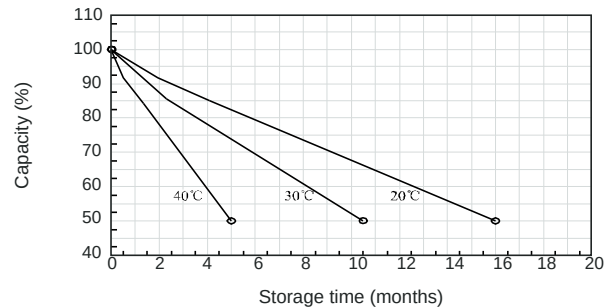
Charging characteristics



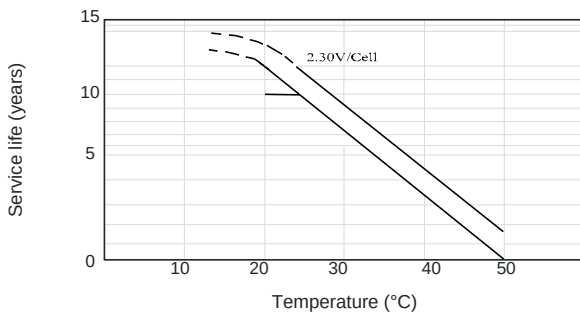
Charge voltage vs. temperature



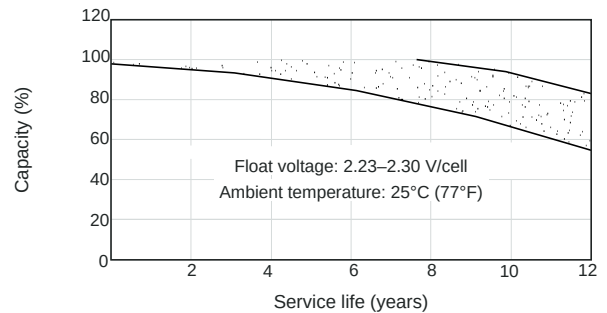
Self-discharge



Float life vs. temperature



Standby service life



Typical characteristic curves. Model-specific ratings are given on pages 1 and 3. Design life is not a warranty.